

1. How Bubble and Selection sorts work

2. Bubble Sort

- a) Explain how the Bubble Sort algorithm works. Discuss its time complexity in the best, worst, and average cases.
- b) Bubble Sort is known to be inefficient. Discuss how optimization can be done in Bubble Sort to improve its performance.
- c) Brief description about the sorting mechanism

3. Selection Sort

- a) Describe the steps involved in the Selection Sort algorithm. What is the time complexity of this algorithm?
- b) Discuss whether Selection Sort is a stable sorting algorithm. How can it be modified to make it stable?
- c) Brief description about the sorting mechanism

4. Merge Sort

- a) Describe the Merge Sort algorithm. Explain the concept of divide and conquer as it applies to Merge Sort.
- c) Brief description about the sorting mechanism

5. Linear vs Binary search operation and complexity

6. Linear Search

- a) Describe how the linear search algorithm works. What is its time complexity in the best, worst, and average cases?
- b) In what scenarios would linear search be the preferred method over other searching algorithms?

7. Binary Search

- a) Explain the binary search algorithm. What assumptions must be made about the input data for binary search to work?
- b) Compare the performance of binary search and linear search for large datasets. In what scenarios would binary search be preferred?

8. Hashing for Searching

- a) Explain how hashing can be used to search for elements in constant time. What are the trade-offs of using hashing compared to other searching methods?

c) Discuss the concepts of collision resolution in hashing (chaining vs. open addressing). How do these methods affect the performance of searching?

9. Search Trees

a) What is a binary search tree (BST)? How does it guarantee efficient searching?

c) Discuss the performance of searching in balanced vs. unbalanced binary search trees. How do AVL trees or red-black trees help improve search efficiency?

10. Tree terminologies

11. roles of data structures and algorithms in today advanced applications

12. examples of ADT used in those applications

13. Why AVL tree?

14. Why BTree?

15. List transition states

16. Advantages and disadvantages of recursion.

17. why worst-case-scenario more relevant?

18. Characteristics and application of a Queue

19. Behavior and operations of DYNAMIC Queue

20. Characteristics and application of a Stack

21. Behavior and operations of a STATIC Stack

22. Concept and common types of an ADT

23. Behavior and operations of SLL and DLL

24. what is pass by value vs pass by reference

25. Structures of Array and Pointers

26. Array vs Pointers advantages and disadvantages

27. Relationship between Data Structures and Algorithms

